Process Characterization Report

A-Mab Drug Substance — Anion Exchange Chromatography (Step 8) — PCR-008

Process Development / Manufacturing Science & Technology (MSAT)

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Abbreviations

ADCC, antibody-dependent cell-mediated cytotoxicity; AEX, anion exchange chromatography; AMV, analytical method validation; CEX, cation exchange chromatography; CPP, critical process parameter; CQA, critical quality attribute; DNA, deoxyribonucleic acid; DoE, design of experiments; DS, drug substance; ELISA, enzyme-linked immunosorbent assay; GPP, general process parameter; HCP, host cell protein; icIEF, imaged capillary isoelectric focusing; ICH, International Council for Harmonisation; IgG1, immunoglobulin G, first subclass; KPP, key process parameter; LOQ, limit of quantitation; LRF, log reduction factor; mAU, milli-absorbance units; MSAT, Manufacturing Science and Technology; MVM, minute virus of mice; NOR, normal operating range; PAR, proven acceptable range; PCP, Process Characterization Plan; PCR, Process Characterization Report; qPCR, quantitative polymerase chain reaction; SD, standard deviation; SOP, standard operating procedure; TCID50, fifty percent tissue culture infectious dose; UV, ultraviolet; WC-CPP, well-controlled critical process parameter; XMuLV, xenotropic murine leukaemia virus.

Executive summary

Anion exchange chromatography (AEX) is the final purification step in the A-Mab drug substance process, and it is operated in flow-through mode so that the antibody passes through the column while acidic impurities bind to the resin. The step sets the cumulative clearance claim for minute virus of mice (MVM). It clears xenotropic murine leukaemia virus (XMuLV), host cell protein (HCP), residual DNA and leached Protein A. This report presents the study defined in PCP-008 as it was executed on a qualified scale-down model, under the lifecycle approach to process validation (U.S. Food and Drug Administration 2011) and the enhanced development approach of ICH Q8, Q9 and Q11 (International Council for Harmonisation 2009, 2023b, 2012). Viral clearance was evaluated within the framework of ICH Q5A (International Council for Harmonisation 2023a). The process model, the quality attribute register and the risk methodology all follow the A-Mab case study (CMC Biotech Working Group 2009).

The study characterized 5 process parameters, of which 4 were studied multivariately and one univariately, and all of them were classified as well-controlled critical process parameters (WC-CPP). Screening over 19 runs identified load pH, equilibration/wash-1 conductivity and load conductivity (factors A, B and C in Table 4.2) as the active factors, whereas protein load had no significant effect on any response across a range that extends above the maximum commercial load. A face-centred central composite design of 28 runs then fitted a response-surface model for each of the 4 measured responses. The screening design identifies which factors matter, and the response-surface model is the predictive model on which the operating region described in §6 is based.

Every governed attribute meets its acceptance criterion across the characterized region. At the worst corner of that region the model predicts an MVM log reduction factor (LRF) of 3.53 $\log_{10}$ against a required step contribution of 3.28 $\log_{10}$. Pool HCP at the corresponding purity corner is predicted at 31.9 ng/mg against an acceptance criterion of 100 ng/mg. Over 2,000 simulated commercial batches the cumulative MVM claim carries a capability index of 1.51, with a predicted mean of 10.38 $\log_{10}$ against a requirement of 8.6 $\log_{10}$, and that is the lowest capability index anywhere in the drug substance process. This step contributes 4.71 $\log_{10}$ of the cumulative MVM figure (47% of the total), and the remainder comes from small-virus retentive filtration (Step 9).

Three deviations were recorded during execution. The most serious of them invalidated the first execution of both designs, because the load material carried an elevated acidic charge variant burden and was not representative of the commercial feed, and both designs were re-executed in full on requalified material. The other two were resolved by correction and by retention with a bounded impact assessment. None of the three altered a parameter classification or a boundary of the operating region, and §13 gives the full account.

The step now provides a defined multivariate operating region over which viral clearance and pool purity both meet their criteria, together with a proven acceptable range (PAR) for every characterized parameter. It does not assure viral safety on its own, since the cumulative MVM claim also rests on small-virus retentive filtration and the cumulative XMuLV claim rests on the low-pH hold as well (PCR-006, PCR-009). No claim is made at this step for glycan variants, charge variants or aggregate. The step's contribution is consolidated with the rest of the train in PCMR-001.

1 Introduction

1.1 Product and unit operation

A-Mab is a humanized IgG1 monoclonal antibody whose principal mechanism of action is antibody-dependent cell-mediated cytotoxicity (ADCC), so the glycan attributes that govern effector function are formed upstream in the production bioreactor (Step 3). The drug substance is purified by a platform train whose steps and roles are listed in Table 1.1.

Table 1.1: The A-Mab drug substance process train and the principal role of each step.

Step	Unit operation	Principal role
3	Production Bioreactor	Forms the glycan, charge-variant and aggregate CQAs (design-space step)
4	Harvest and Clarification	Primary recovery and clarification; forms no product-quality CQA
5	Protein A Chromatography	Capture; sets leached Protein A; principal HCP and DNA clearance
6	Low-pH Viral Inactivation	Low-pH hold; sets the cumulative XMuLV (enveloped) clearance
7	Cation Exchange Chromatography	Polish; principal aggregate reduction; major HCP/DNA/leached-PA clearance
8	Anion Exchange Chromatography	Flow-through polish; sets the cumulative MVM clearance; clears XMuLV/HCP/DNA
9	Small-Virus Retentive Filtration	Dedicated small-virus removal; principal MVM clearance
10	Ultrafiltration / Diafiltration	Formulation / mass balance; delivers drug substance at target concentration

Anion exchange chromatography is the last purification step in that train. It receives the cation exchange (CEX) pool after adjustment to the target pH and conductivity. The column is operated in flow-through mode, which means that at the load pH the antibody carries a net positive charge and passes through the strong anion exchanger while acidic impurities bind and are retained. Product is collected across the load and the equilibration wash on ultraviolet (UV) absorbance. Because the mechanism is adsorptive for the impurities and non-adsorptive for the product, the column is not loaded to a product breakthrough limit, and the step tolerates a high protein load.

This is the step at which the cumulative clearance claim for MVM is set. Charge-based partitioning of a small non-enveloped parvovirus is efficient at the operating pH, and the mechanism is independent of the size-based mechanism used at the small-virus retentive filter (Step 9, PCR-009), which is what allows the two contributions to be added. The step also clears XMuLV, HCP, residual DNA and leached Protein A, and at the nominal commercial operating point it reduces HCP from 117 ng/mg in the load to 19 ng/mg in the pool, a clearance factor of 6.2, at a step yield of about 97.3%. No claim is made at this step for glycan variants, charge variants or aggregate, because those attributes are formed at Step 3 and controlled at Step 3 and Step 7, and anion exchange does not discriminate between them under platform conditions.

1.2 Objectives

The study had five objectives.

- (i) Quantify the relationship between each characterized process parameter and each measured response, over ranges deliberately wider than the intended commercial control ranges.
- (ii) Define a multivariate operating region over which the governed quality attributes remain within their acceptance criteria.
- (iii) Establish a proven acceptable range for each characterized parameter, both at set-point and with the remaining parameters varying within their normal operating ranges (NOR).
- (iv) Classify each process parameter against the decision criteria in SOP-4001.
- (v) Justify the Stage 2 operating ranges and the step's contribution to the control strategy for transfer to the commercial facility.

1.3 Regulatory and scientific basis

The study is a Stage 1 process design activity under the lifecycle approach to process validation (U.S. Food and Drug Administration 2011). It follows the enhanced development approach of ICH Q8 together with the risk management principles of ICH Q9, which are applied through the criticality framework of §2.2 (International Council for Harmonisation 2009, 2023b). Criticality is treated as a continuum and not as a binary label, and attributes of high or very high criticality are called critical. ICH Q11 provides the basis for linking parameter ranges to a design space for the drug substance (International Council for Harmonisation 2012). The viral clearance work follows ICH Q5A (International Council for Harmonisation 2023a), under which clearance is claimed on a modular basis, with each step evaluated independently in a qualified small-scale spiking model. The process model, the quality attribute register and the risk methodology are all taken from the published A-Mab case study (CMC Biotech Working Group 2009).

1.4 Related documents

The documents that scope this study, and the document into which it rolls up, are listed in Table 1.2. The paired protocol is PCP-008 and the risk basis is RA-001.

Table 1.2: Related documents in the A-Mab characterization package.

Document ID	Title	Relationship
PTP-001	Process Transfer Plan (A-Mab Drug Substance)	Parent — transfer scope
RA-001	Pre-Characterization Process Risk Assessment (A-Mab Drug Substance)	Parent — risk basis for study scope
PCMP-001	Process Characterization Master Plan (A-Mab Drug Substance)	Parent — master plan
PCP-008	Process Characterization Plan (Anion Exchange Chromatography (Step 8))	Sibling — paired document
PCMR-001	Process Characterization Master Report (A-Mab Drug Substance)	Rolls up into

2 Prior knowledge and quality risk basis

Prior knowledge and the risk assessment recorded in RA-001 together set the factor list and the ranges used in this study. The platform behaviour of a flow-through anion exchange step is well established for this antibody class, so the work described here confirms and bounds that behaviour for A-Mab and is not a discovery study. The mechanistic expectation is therefore given first, then the attributes in scope, and then the parameter selection that follows from both.

2.1 Platform and prior-product knowledge

The A-Mab downstream process is a platform train with an extensive performance history. The same anion exchange polishing step has been operated for three previous humanized IgG1 antibodies (X-Mab, Y-Mab and Z-Mab) and throughout A-Mab clinical and engineering manufacture, and that experience establishes three expectations against which the present results can be tested.

The first expectation concerns viral clearance. Removal of both model viruses on a strong anion exchanger is a charge partitioning phenomenon, so clearance falls as the load pH falls, because the virus carries less negative charge and binds the resin less strongly, and it falls again as the load conductivity rises, because counter-ions compete for the same binding sites. Load pH and load conductivity were therefore expected to dominate both viral responses, and to act in the same direction on each of them.

The second expectation concerns HCP. Host cell protein is a heterogeneous population with a wide range of isoelectric points, so its retention depends on the ionic environment of the column during the flow-through and wash phases, and that environment is set by the conductivity of the equilibration and wash-1 buffer. Prior platform data show that a higher wash conductivity weakens the binding of the more weakly acidic species and releases them into the pool, whereas a higher load pH has the opposite effect because it increases the net negative charge of those same species. An interaction between the two was anticipated on that basis.

The third expectation concerns capacity. Since the product does not bind, the column is loaded to an impurity breakthrough limit and not to a product breakthrough limit, and platform experience with the prior products places that limit well above the intended commercial load. Protein load was expected to be inactive over a wide range. The study was designed to test that expectation and not to assume it.

Prior knowledge also fixes what this step is not expected to do. Glycosylation variants are not separated by charge-based chromatography, apart from sialylated species, which are present at very low levels in A-Mab, and aggregate is polished at cation exchange (Step 7). The prior knowledge transfer is limited to mechanism and to the direction of effects. Every quantitative claim in this report rests on the data generated in the executed study.

2.2 Quality attributes in scope

The step sets one critical quality attribute (CQA), which is given in Table 2.1 with its acceptance criterion and with its criticality score under Tool #1 of the A-Mab framework, a score that combines the impact of the attribute with the uncertainty around it.

Table 2.1: Critical quality attribute set by the anion exchange step.

CQA	Category	Acceptance	Criticality	Tool #1
Viral clearance — MVM (parvovirus)	viral safety	8.6–20 log10 (cumulative)	VH	20

MVM clearance is of very high criticality, and its acceptance criterion is cumulative across the process and expressed as a log reduction factor, with a requirement of 8.6 $\log_{10}$ over the whole train. Anion exchange and small-virus retentive filtration are the only two steps credited with MVM clearance, so subtracting the nominal contribution of Step 9 from the cumulative requirement gives the contribution that this step must deliver, which is 3.28 $\log_{10}$. That back-calculated figure is the step-level acceptance criterion used throughout this report. The same construction applied to XMuLV gives a step-level criterion of 3.78 $\log_{10}$.

The step also clears four attributes that it does not set (Table 2.2).

Table 2.2: Quality attributes cleared but not set by the anion exchange step.

CQA	Category	Acceptance	Criticality	Tool #1
Host Cell Protein (HCP)	process impurity	0–100 ng/mg	M-H	36
Residual DNA	process impurity	0–0.001 ng/dose	VL	6
Leached Protein A	process impurity	0–5 ppm	L-M	16
Viral clearance — XMuLV (enveloped)	viral safety	16.7–30 log10 (cumulative)	VH	20

Host cell protein is the attribute in this group with any prospect of constraining the operating region, because it is of moderate to high criticality and anion exchange is the last step that clears it. Neither virus filtration nor ultrafiltration and diafiltration (Step

10) removes HCP, so the concentration in the anion exchange pool carries through to the drug substance (DS), and the drug substance specification is applied directly to the pool. Residual DNA and leached Protein A are cleared with large margins and are reported here for completeness, the step being credited with 4.0 $\log_{10}$ of DNA clearance and 2.0 $\log_{10}$ of leached Protein A clearance at the nominal operating point.

2.3 Risk-based prioritization and parameter selection

RA-001 assessed every operating parameter of the step against its potential impact on quality attributes and against its potential to interact with other parameters. Each parameter was then placed into one of three groups. The first holds parameters warranting multivariate evaluation, and the second holds secondary parameters whose ranges could be supported by univariate study. The third holds parameters needing no new study, because their ranges follow from platform knowledge or from a modular claim established on the prior products. Where no data or rationale were available to support an assessment, the parameter was ranked at the highest level.

The first group holds 4 parameters. Load pH and load conductivity were ranked high because prior knowledge links both of them directly to viral clearance, and equilibration and wash-1 conductivity was ranked high because it governs the ionic environment during flow-through and because an interaction with load pH was anticipated. Protein load was ranked high on interaction potential and not on main effect, since an impurity breakthrough limit would present itself as a load interaction before it presented itself as a load main effect.

Operating flow rate fell into the second group, because flow rate acts through residence time, and residence time affects the approach to adsorption equilibrium and not the position of that equilibrium. An interaction with the chemistry factors was not expected on that basis, so a univariate assessment was judged sufficient.

Column bed height, resin lot, cleaning and regeneration conditions and the buffer salt system fell into the third group. Bed height and linear velocity are fixed by the scale-down convention (§3.1), resin life-cycle behaviour is controlled under SOP-2008 and is supported by the platform data on the prior products, and the buffer salt system is fixed by SOP-2011.

The characterized ranges were deliberately set wider than the intended commercial control ranges (the NOR), each range being extended to roughly two or three times the routine control range so that process performance could be assessed well beyond the conditions the commercial step will encounter. The wider ranges also supply the process understanding needed to support later movement within the design space. The individual ranges and their justification are given in §4.1.

3 Materials and methods

3.1 Scale-down model and its qualification

A laboratory-scale chromatography system was qualified as a model of the commercial step under SOP-1001, and it was built on the standard scaling convention for column chromatography. Column diameter was reduced, while bed height, linear velocity, protein load per litre of resin and load volume expressed in column volumes were all held equal to the commercial values. This convention holds residence time and mass transport constant across scales, and all buffer volumes are expressed in column volumes, so the same proportional quantities are used at both scales.

Qualification compared the laboratory model with commercial-scale batch data on the input and output attributes of the step (step yield, pool HCP, pool acidic variants and pool aggregate), and column efficiency was compared by the usual measures (plate count and peak asymmetry). No statistically significant difference between scales was found for any of these attributes. The qualification record is held in the scale-down model qualification report referenced by PCP-008. On that basis the model is considered representative of the commercial step, and the operating range claims derived from it are applied to commercial scale.

Two limits on the scale-down warrant are recorded here. The model reproduces the commercial step at the conditions at which it was qualified, which are the target conditions, and it is not independently qualified at the edges of the characterized ranges. It also uses a single packed column and a single resin lot within a study, so it carries no information about resin lot variability or about end-of-lifetime resin performance, both of which are controlled under SOP-2008 and are addressed in the transfer programme described in PTP-001.

Viral clearance was measured in a small-scale spiking model derived from the same system, in accordance with ICH Q5A (International Council for Harmonisation 2023a). Each spiking condition was run in duplicate and the lower of the two log reduction factors was carried forward. The spiking runs were executed under conditions chosen to be conservative for clearance: flow rate and protein load were set at or above the maximum values used in the characterization runs, and bed height and residence time were set at their minimum operating values. The collection criteria were set to include both the earliest and the latest eluting material.

The centre-point replicates within each design provide the pure-error estimate used in the lack-of-fit test (§5.3). This is a within-study estimate of reproducibility, and it carries both process and assay variation.

3.2 Operation

Runs were executed under the step operating procedure (SOP-2011). The cation exchange pool was adjusted to the target load pH and load conductivity with titrant and dilution buffer and held at ambient temperature until loading. The column was equilibrated with the equilibration and wash-1 buffer at the conductivity set for the run, and the load was then applied at the run flow rate. The column was washed with equilibration buffer, and product was collected across the load and the wash on UV absorbance at 280 nm, with collection started and stopped on the ascending and descending absorbance thresholds specified in SOP-2011. A deviation on the descending threshold is recorded below (§13.2).

Column packing, sanitization, storage and life-cycle control followed SOP-2008, and bed height was held at the commercial value in every run. Buffer preparation and release followed SOP-2103. Inputs that are controlled to platform specification (resin type, buffer salt system and column hardware) were held at their specified values and were not varied, because they are identity and quality controlled and are not characterization variables.

3.3 Analytical methods

Every response was measured by a validated method, and the controlled procedures and method validations that apply to this step are listed in Table 3.1.

Table 3.1: Controlled procedures and validated analytical methods for the step.

Reference	Title	Type
SOP-1001	Qualification of Scale-Down Models	SOP
SOP-1002	Design, Execution and Statistical Analysis of DoE Studies	SOP
SOP-2011	Operation of the Anion-Exchange Polishing Chromatography Step	SOP
SOP-2008	Chromatography Resin Life-Cycle, Packing and Sanitization	SOP
SOP-4001	Process-Parameter Classification and Control-Strategy Definition	SOP
AMV-3012	Host-Cell Protein ELISA	Method validation
AMV-3014	Residual DNA (qPCR)	Method validation
AMV-3016	Leached Protein A by ELISA (ppm)	Method validation
AMV-3017	XMuLV Infectivity Titre (TCID50)	Method validation
AMV-3018	MVM Infectivity Titre (TCID50/qPCR)	Method validation

Reference	Title	Type
AMV-3013	Charge Variants (icIEF)	Method validation

Pool HCP was measured by the process-specific enzyme-linked immunosorbent assay validated under AMV-3012, residual DNA by quantitative polymerase chain reaction (AMV-3014) and leached Protein A by enzyme-linked immunosorbent assay (AMV-3016), while infectivity titres for XMuLV and MVM were determined under AMV-3017 and AMV-3018 respectively. Charge variants were measured by imaged capillary isoelectric focusing (AMV-3013), which was applied to the load material and not to the pool, because the step makes no charge variant claim. That method is reported here because it identified the root cause of the first recorded deviation (§13.1).

Assay performance matters to the interpretation of the results, and the relevant figures are tabulated in Appendix D. The HCP assay is the least precise of the methods used, with an intermediate precision of 9.5% relative standard deviation and about 40% of the observed variance in a pool result attributable to the assay itself. The infectivity assays are considerably more precise on the $\log_{10}$ scale, at 0.32 $\log_{10}$ for MVM and 0.30 $\log_{10}$ for XMuLV. Variance in the HCP results is attributed to the assay wherever the magnitude is consistent with these figures. That attribution is made explicit in the results (§5.1).

3.4 Sampling plan

One pool sample was drawn from each run and assayed for HCP, residual DNA and leached Protein A, and step yield was calculated from load and pool protein mass determined by UV absorbance at 280 nm. Virus-spiked runs were sampled at the load and across the collected pool, and titres were determined in duplicate. Load material was sampled before each execution and tested for charge variants, HCP and aggregate so that feed representativeness could be verified before the design was run, and it was that verification which detected the first of the recorded deviations (§13.1).

The screening design carried 3 centre-point runs and the response-surface design carried 4. Centre points were distributed through the run order (never consecutively), and run order was randomized within each design.

3.5 Statistical methods

All factors were analysed in coded units, where the coded levels −1/0/+1 correspond to the low edge, the set-point and the high edge of each characterized range. Coding places the factors on a common scale, so that effect estimates are directly comparable across parameters measured in different units.

The screening data were fitted by ordinary least squares (SOP-1002) with all main effects and all two-factor interactions, and an effect is reported as twice the coded coefficient, which is the change in the response across the full characterized range of that factor (the coded levels −1 and +1). The response-surface data were fitted with the full quadratic model, which adds the squared terms to the same structure. Significance was assessed at $\alpha = 0.05$ throughout.

Model adequacy was judged on four criteria. The first two are the coefficient of determination and its adjusted form, together with the predicted coefficient of determination computed from the prediction sum of squares. The third is the overall model F test, and the fourth is the lack-of-fit test that compares the residual variation not explained by the model with the pure error estimated from the centre-point replicates. A model is treated as adequate when the F test is significant and the lack-of-fit test is not. A factor is treated as inactive when it appears in no significant term.

Commercial-scale capability was estimated by Monte-Carlo simulation of 2,000 batches, in which each parameter was drawn within its NOR, the fitted models were evaluated at the drawn conditions, and process and assay variation were added. Capability is reported as a one-sided process capability index against the relevant acceptance limit. The proven acceptable ranges were computed by the two analyses set out below (§7).

4 Study design

The design has two stages: a screening design that resolves which factors are active and which interactions exist, and a response-surface design that quantifies the surface over those factors and provides the predictive model. A univariate assessment covers the one parameter that was not carried into the multivariate work, and together the three cover the knowledge space defined below.

4.1 Factors, ranges and the knowledge space

The characterized parameters are given in Table 4.1 with their set-points, their normal operating ranges, the ranges studied and their final classification, which is an outcome of the study and is justified below (§9).

Table 4.1: Characterized process parameters, ranges studied and final classification.

Parameter	Unit	Set-point	NOR	Char. range	Class	Study
Load pH	pH	7.5	7.35–7.65	7.2–7.8	WC-CPP	multivariate
Equil/Wash-1 conductivity	mS/cm	2.6	2.1–3.1	1.6–3.6	WC-CPP	multivariate
Load conductivity	mS/cm	5.5	4–7	3–8	WC-CPP	multivariate
Protein load	g/L resin	200	120–280	50–300	WC-CPP	multivariate
Operating flow rate	cm/hr	300	200–400	150–450	WC-CPP	univariate

The characterization range is the knowledge space edge for each parameter and is not a proven acceptable range, since the PAR is a computed, per-attribute quantity (§7). The normal operating range sits inside the characterization range in every case.

Load pH was studied from 7.2 to 7.8, where the lower edge was set by the viral clearance mechanism, because clearance of both model viruses falls as pH falls. The upper edge was set by the need to keep the antibody comfortably away from its isoelectric point so that it continues to flow through. Equilibration and wash-1 conductivity was studied from 1.6 to 3.6 mS/cm, the upper edge being set by prior platform data showing loss of HCP retention above that value. Load conductivity was studied from 3 to 8 mS/cm, which spans the conductivity that the cation exchange pool can present after adjustment. Protein load was studied from 50 to 300 g/L resin, with the upper edge set above the maximum load the commercial step will apply so that any impurity breakthrough limit would fall inside the studied range. Operating flow rate was studied from 150 to 450 cm/hr, which brackets the commercial linear velocity by a wide margin in both directions.

The letter code used for the factors in the effect and coefficient tables is given in Table 4.2.

Table 4.2: Coded factor legend used in the effect and coefficient tables.

Code	Factor	Unit	Studied in
A	Load pH	pH	screening + RSM
B	Equil/Wash-1 conductivity	mS/cm	screening + RSM
C	Load conductivity	mS/cm	screening + RSM
D	Protein load	g/L resin	screening + RSM

The designs measured 4 responses on every run. Pool HCP, the XMuLV log reduction factor and the MVM log reduction factor are the quality responses. Step yield is a process performance response and is not linked to a quality attribute.

4.2 Screening design

The screening design was a two-level full factorial in the 4 multivariate factors augmented with 3 centre points, comprising 19 runs in total of which 16 are factorial. The design estimates all main effects and all two-factor interactions (six of them) without aliasing. The centre points supply an estimate of pure error together with a test for curvature. The full coded design matrix with its responses is given in Appendix A.

4.3 Response-surface design

The response-surface design was a face-centred central composite design in the same 4 factors, comprising 28 runs made up of 16 factorial runs, 8 axial runs and 4 centre points. The axial points sit on the faces of the cube, so the design stays inside the characterized ranges and requires no setting beyond them, and it supports the full quadratic model, so it resolves curvature in each factor as well as the two-factor interactions. This design was executed on requalified load material, and the full coded design matrix with its responses is given in Appendix B.

4.4 Univariate assessment

Operating flow rate was assessed one factor at a time across its characterized range with all other parameters held at set-point. It was excluded from the multivariate designs on mechanistic grounds, because it acts on the rate of approach to adsorption equilibrium and not on the position of that equilibrium, so an interaction with pH or conductivity was not expected. The viral clearance spiking runs were executed at the maximum flow rate, which is the conservative condition for a rate-limited mechanism, and clearance was maintained, so flow rate is bounded by direct evidence at the edge of its range for the response most sensitive to residence time.

5 Results

5.1 Centre-point performance and reproducibility

Reproducibility at the centre point is good for the viral clearance and yield responses and only moderate for HCP. The centre-point mean, standard deviation and coefficient of variation for each response in the response-surface design are given in Table 5.1.

Table 5.1: Centre-point reproducibility in the response-surface design.

Response	n	Mean	SD	%CV
Pool HCP (ng/mg)	4	13.7	2.48	18.2
XMuLV LRF ($\log_{10}$)	4	6.63	0.445	6.71
MVM LRF ($\log_{10}$)	4	4.9	0.336	6.86
Step yield	4	0.966	0.0123	1.27

Pool HCP is the least precise of the four responses, at 18.2% coefficient of variation over 4 centre points, against 20.6% for the same response in the screening design. The intermediate precision of the HCP assay alone is 9.5% relative standard deviation, and about 40% of the variance of a pool result is attributed to the assay (AMV-3012), so most of the observed centre-point scatter in this response is analytical and not processing variation. The two log reduction factor responses reproduce at 6.9% and 6.7%, and step yield at 1.27%.

These figures set the resolution of the study. An effect on pool HCP smaller than a few ng/mg cannot be separated from assay noise in a design of this size. The pure error term derived from these replicates is used in the lack-of-fit tests reported in §5.3.

5.2 Screening: factor effects

The screening design resolves the factor set to three active factors (A, B and C), and the model summary for each response is given in Table 5.2.

Table 5.2: Screening model summary for each response.

Response	N	R^2	Adj. R^2	Pred. R^2	F	p-value	RMSE
Pool HCP (ng/mg)	19	0.966	0.923	0.841	22.6	8.55e-05	2.45

Response	N	R^2	Adj. R^2	Pred. R^2	F	p-value	RMSE
XMuLV LRF ($\log_{10}$)	19	0.842	0.644	-0.106	4.25	0.0257	0.393
MVM LRF ($\log_{10}$)	19	0.97	0.932	0.869	25.8	5.18e-05	0.178
Step yield	19	0.498	-0.13	-3.64	0.792	0.642	0.0126

Pool HCP and MVM clearance are well described by the screening model, at R^2 = 0.966 and 0.970. XMuLV clearance is described less well at 0.842 and has a negative predicted R^2. Step yield is not described by the model at all (F test p = 0.642). The screening model is near-saturated, with 10 terms fitted from 19 runs, so these fit statistics identify structure without supporting prediction, and the response-surface models are the predictive models (§5.3).

Pool HCP is governed by two factors and by their interaction, and the six largest effects are given in Table 5.3.

Table 5.3: Largest screening effects on pool host cell protein.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
B	12.1	6.04	0.613	9.85	9.48e-06	***
A	-11.9	-5.96	0.613	-9.72	1.05e-05	***
AB	-6.63	-3.32	0.613	-5.41	0.000642	***
C	-1.66	-0.829	0.613	-1.35	0.213	
AC	1.65	0.827	0.613	1.35	0.214	
BC	-1.21	-0.605	0.613	-0.986	0.353	

Equilibration and wash-1 conductivity raises pool HCP by 12.1 ng/mg across its characterized range, load pH lowers it by 11.9 ng/mg across its own range, and both effects are significant at $p < 0.001$. The interaction between them is also significant, at -6.6 ng/mg with $p < 0.001$, and its sign means that the penalty for a high wash conductivity is smaller when the load pH is high. Load conductivity and protein load are not significant for this response.

MVM clearance is governed by load pH and load conductivity, whose effects are given with the four next largest terms in Table 5.4.

Table 5.4: Largest screening effects on MVM log reduction factor.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
A	1.22	0.61	0.0446	13.7	7.83e-07	***
C	-0.715	-0.357	0.0446	-8.01	4.32e-05	***
B	0.182	0.091	0.0446	2.04	0.0757	
BD	0.107	0.0536	0.0446	1.2	0.263	
AD	-0.0627	-0.0314	0.0446	-0.704	0.502	

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
D	-0.0451	-0.0226	0.0446	-0.506	0.626	

Load pH raises the MVM log reduction factor by 1.22 $\log_{10}$ across its range and load conductivity lowers it by 0.71 $\log_{10}$ across its own, and both are significant at $p < 0.001$. No interaction term is significant for this response, and equilibration and wash-1 conductivity gives a small positive estimate that does not reach significance ($p = 0.076$), which is treated here as a null result.

XMuLV clearance follows the same pattern with smaller magnitudes (Table 5.5).

Table 5.5: Largest screening effects on XMuLV log reduction factor.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
A	1	0.501	0.0982	5.1	0.000934	***
C	-0.605	-0.302	0.0982	-3.08	0.0151	*
D	0.268	0.134	0.0982	1.37	0.209	
CD	-0.255	-0.128	0.0982	-1.3	0.23	
AD	0.25	0.125	0.0982	1.27	0.24	
B	0.198	0.0991	0.0982	1.01	0.342	

Load pH raises the XMuLV log reduction factor by 1.00 $\log_{10}$ across its range ($p < 0.001$) and load conductivity lowers it by 0.60 $\log_{10}$ ($p = 0.015$), while no other term reaches significance. The direction of both effects matches the direction seen for MVM, which is the behaviour that the charge partitioning mechanism predicts.

Protein load is inactive on every response, its largest estimate being 0.89 ng/mg on pool HCP ($p = 0.491$). No interaction involving protein load is significant. This is a null result over a range reaching 300 g/L resin, which lies above the maximum commercial load, so protein load is retained in the knowledge space as evidence that the step is robust to it and is classified on that basis in §9. No impurity breakthrough limit was found inside the studied range.

Step yield is not affected by any factor, since no term reaches significance and the model F test is not significant. Because the centre-point coefficient of variation for yield is 1.27%, the response was measured precisely enough for a real effect of any practical size to have been detected, and a flow-through step in which the product does not bind is expected to behave in exactly this way.

5.3 Response-surface models

The response-surface models are adequate for the three quality responses and are the basis of the operating region (§6). The model summary is given in Table 5.6.

Table 5.6: Response-surface model summary for each response.

Response	N	R^2	Adj. R^2	Pred. R^2	F	p-value	RMSE
Pool HCP (ng/mg)	28	0.934	0.864	0.66	13.2	1.83e-05	2.65
XMuLV LRF ($\log_{10}$)	28	0.915	0.824	0.621	10	8.69e-05	0.314
MVM LRF ($\log_{10}$)	28	0.898	0.787	0.394	8.14	0.000269	0.32
Step yield	28	0.488	-0.0632	-2.18	0.885	0.589	0.0145

The three quality responses give R^2 between 0.898 and 0.934 with adjusted R^2 of at least 0.787, and each of them has a significant model F test. Predicted R^2 is lowest for MVM clearance at 0.39, which reflects the number of terms carried in the full quadratic model relative to 28 runs. Step yield again has no significant model ($p = 0.589$), so no predictive model for yield is claimed here.

The lack-of-fit test supports the model form for each quality response, and the partition for MVM clearance is given in Table 5.7.

Table 5.7: Analysis of variance with lack-of-fit partition for MVM clearance.

Source	Sum sq.	df	Mean sq.	F	p-value
Model	11.7	14	0.834	8.14	0.000269
Residual	1.33	13	0.102	nan	nan
Lack of fit	0.994	10	0.0994	0.881	0.619
Pure error	0.339	3	0.113	nan	nan

Lack of fit is not significant for MVM clearance, at $F = 0.88$ with $p = 0.619$, and the same test is not significant for pool HCP ($p = 0.499$) or for XMuLV clearance ($p = 0.911$). The corresponding partition for pool HCP is given in Table 5.8.

Table 5.8: Analysis of variance with lack-of-fit partition for pool host cell protein.

Source	Sum sq.	df	Mean sq.	F	p-value
Model	1.3e+03	14	93.1	13.2	1.83e-05
Residual	91.5	13	7.04	nan	nan
Lack of fit	73	10	7.3	1.19	0.499
Pure error	18.5	3	6.16	nan	nan

The full coefficient table for MVM clearance is given in Table 5.9.

Table 5.9: Response-surface coefficients for MVM log reduction factor.

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	4.85	0.111	43.9	1.63e-15	***
A	0.539	0.0754	7.15	7.48e-06	***
B	0.0307	0.0754	0.407	0.691	
C	-0.564	0.0754	-7.48	4.65e-06	***
D	0.0261	0.0754	0.346	0.735	
AB	0.127	0.08	1.59	0.136	
AC	-0.074	0.08	-0.925	0.372	
AD	0.0353	0.08	0.441	0.666	
BC	-0.0339	0.08	-0.424	0.679	
BD	0.00805	0.08	0.101	0.921	
CD	-0.0156	0.08	-0.195	0.848	
A^2	-0.322	0.199	-1.61	0.13	
B^2	0.149	0.199	0.747	0.469	
C^2	0.0224	0.199	0.112	0.912	
D^2	0.135	0.199	0.679	0.509	

Only the two main effects are significant, at 0.539 $\log_{10}$ per coded unit for load pH and -0.564 $\log_{10}$ per coded unit for load conductivity, and no interaction term and no quadratic term reaches significance. The largest quadratic estimate is on load pH, at -0.322 with $p = 0.130$, which suggests a slight flattening of the surface at high pH without providing evidence for it. The MVM surface is close to planar over the region studied, and the two governing parameters act additively on it.

The full coefficient table for pool HCP is given in Table 5.10.

Table 5.10: Response-surface coefficients for pool host cell protein.

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	14.8	0.917	16.2	5.51e-10	***
A	-5.34	0.625	-8.54	1.09e-06	***
B	6.19	0.625	9.89	2.04e-07	***
C	0.0399	0.625	0.0637	0.95	
D	0.387	0.625	0.619	0.546	
AB	-1.45	0.663	-2.19	0.0475	*
AC	-0.925	0.663	-1.39	0.187	
AD	0.0192	0.663	0.029	0.977	
BC	0.15	0.663	0.226	0.825	
BD	-0.559	0.663	-0.843	0.415	
CD	-0.124	0.663	-0.187	0.855	
A^2	1.91	1.65	1.16	0.268	
B^2	2.22	1.65	1.34	0.202	
C^2	-0.0187	1.65	-0.0113	0.991	
D^2	-1.98	1.65	-1.2	0.253	

Equilibration and wash-1 conductivity and load pH are again the significant main effects, at 6.19 and -5.34 ng/mg per coded unit, and their interaction is significant ($p = 0.047$) and is retained in the model. The screening estimate of that interaction was the larger of the two, and the response-surface value is the one carried forward, because it is estimated with curvature in the model and on a design that supports it. No quadratic term is significant for pool HCP either.

The fitted surfaces over load pH and equilibration and wash-1 conductivity, with the remaining factors at set-point, are shown in Figure 5.1.

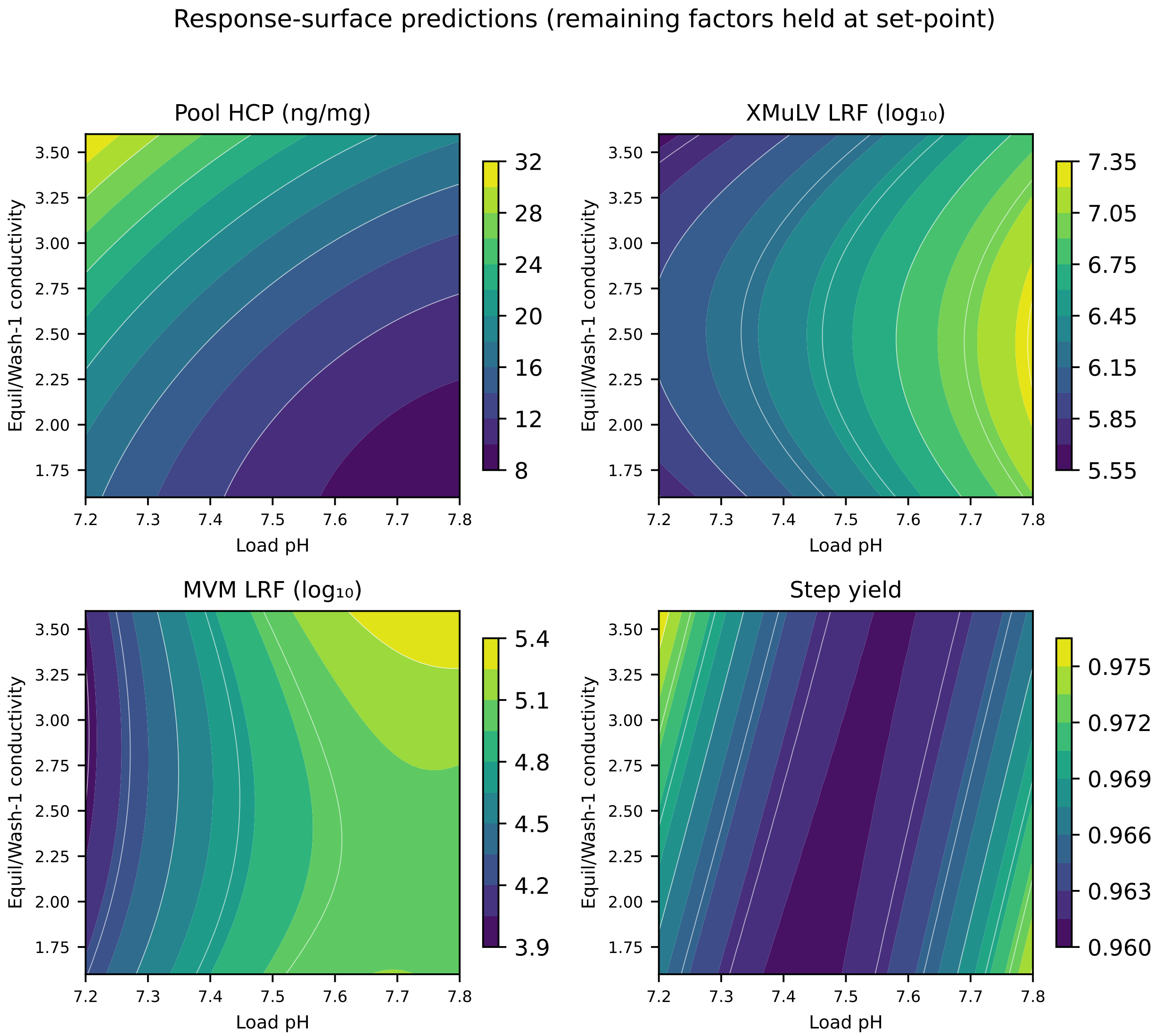

Figure 5.1: Response-surface predictions over load pH and equilibration/wash-1 conductivity, with load conductivity and protein load held at set-point.

The HCP panel shows the interaction directly, since the contours converge towards high load pH and the pool concentration becomes progressively less sensitive to wash conductivity as pH rises. The two viral clearance panels are almost flat in the wash conductivity direction. That is the graphical form of the null result for that factor on clearance. The same responses are shown over load pH and load conductivity in Figure 5.2.

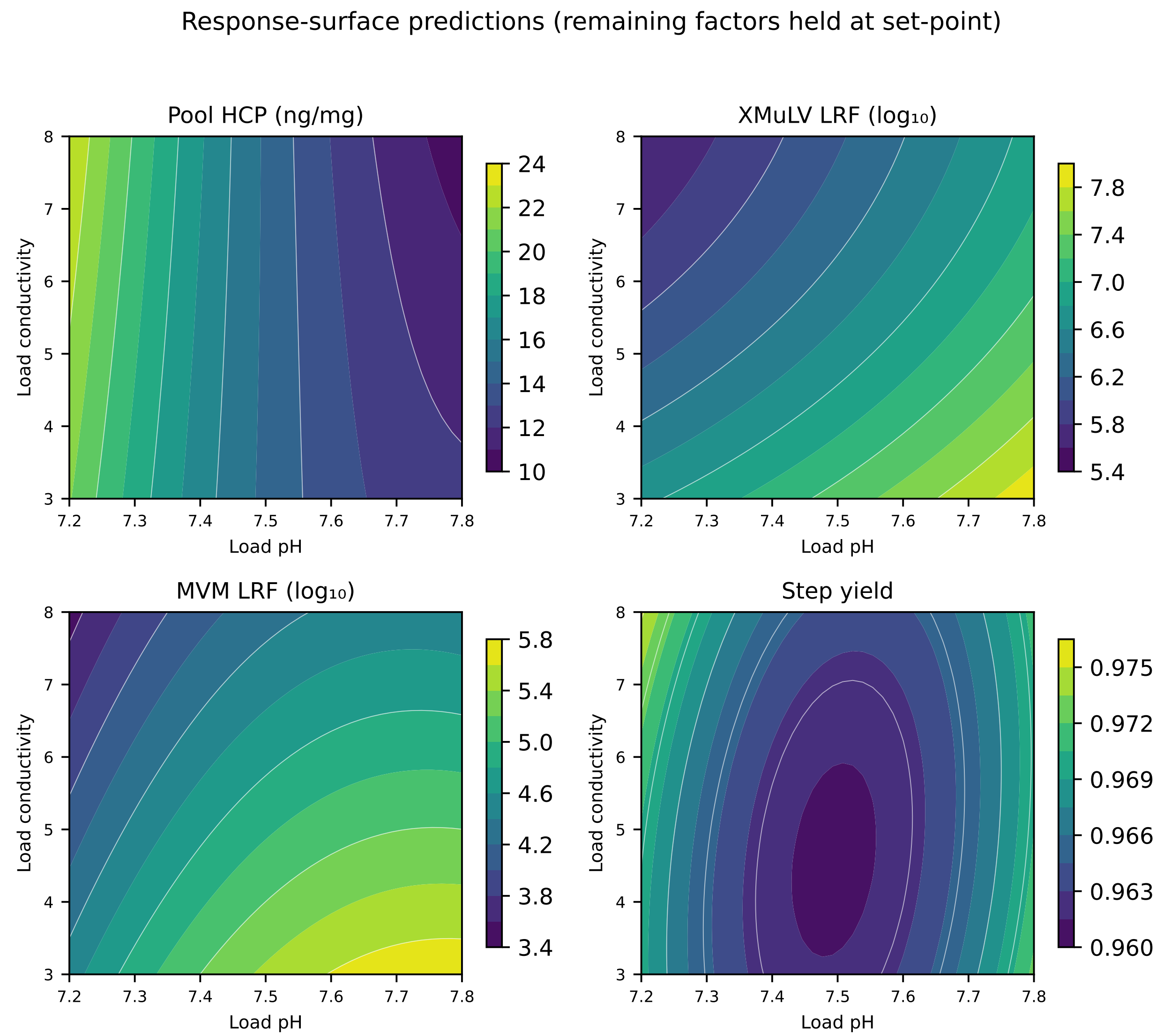

Figure 5.2: Response-surface predictions over load pH and load conductivity, with equilibration/wash-1 conductivity and protein load held at set-point.

Both viral clearance panels fall along the same diagonal, with clearance highest at high load pH and low load conductivity and lowest at low load pH and high load conductivity, and that diagonal is the plane on which the operating region is defined (§6).

Model diagnostics support the fitted models, and those for MVM clearance are shown in Figure 5.3.

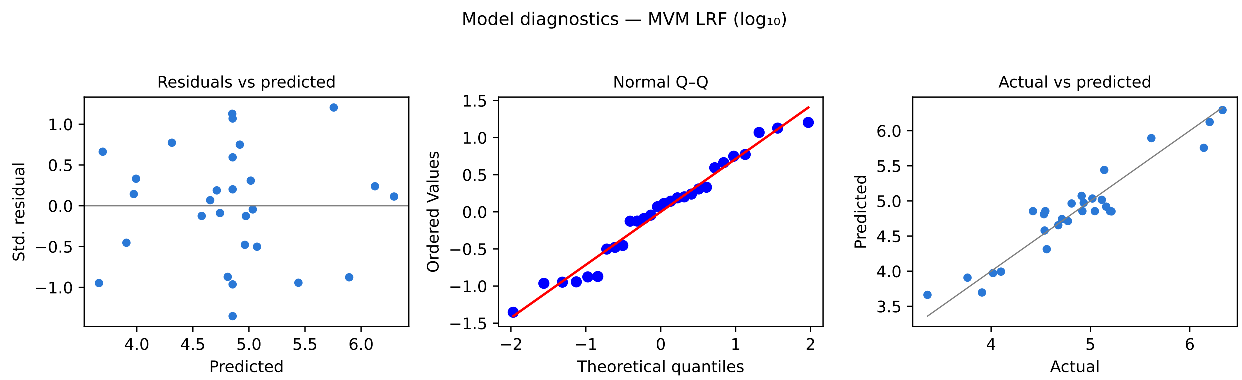

Figure 5.3: Model diagnostics for the MVM log reduction factor response-surface model.

The residuals show no structure against the predicted value, the normal quantile plot is close to linear, and the actual against predicted plot lies on the identity line across the range of the response. The same three diagnostics for pool HCP are shown in Figure 5.4.

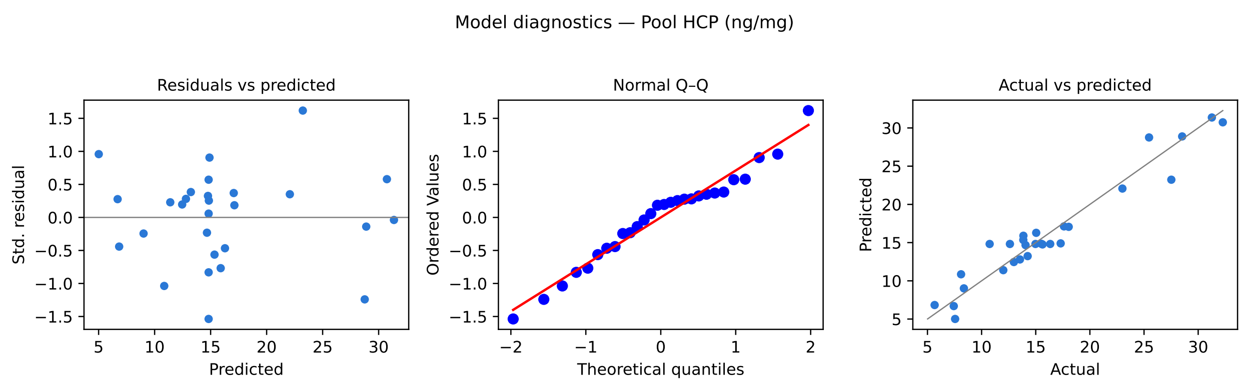

Figure 5.4: Model diagnostics for the pool host cell protein response-surface model.

The HCP residuals are larger in absolute terms and increase mildly with the predicted value, which is the pattern expected for a concentration measured by an assay of constant relative standard deviation. It does not indicate a defect in the model form, and the lack-of-fit test for this response is not significant.

5.4 Mechanistic interpretation

The surfaces confirm the mechanistic expectation set out in §2.1, and that confirmation is what allows the operating region to be defined multivariately in §6 instead of parameter by parameter.

Viral clearance behaves as a charge partitioning process, in which both model viruses are more strongly retained when they carry more negative charge and when fewer competing counter-ions are present. Raising the load pH achieves the first and lowering the load conductivity achieves the second. The fitted coefficients for both viruses carry the signs that those statements require. Because no interaction between pH and conductivity is significant in the MVM model, the two act additively over the region studied. A loss of clearance caused by a low load pH can be offset by a lower load conductivity, and the amount of offset required is given by the ratio of the two coefficients.

Host cell protein behaves differently, because it is a mixture and not a single species. Its governing parameter is the conductivity of the equilibration and wash-1 buffer and not the conductivity of the load. That says that the binding of the weakly retained species is decided by the ionic environment in which the column is held during flow-through and wash, and that the ionic strength of the load itself contributes little once the load has been adjusted. The load pH interaction follows from the same picture, since at a higher pH the weakly acidic species carry more negative charge and remain bound even when the wash conductivity is raised, which is why the two sets of contours converge.

Protein load is inactive across a range that extends above the commercial load, which is consistent with a flow-through step in which the product does not compete for binding sites and the impurity load remains far below the capacity of the resin. It also means that the impurity breakthrough limit lies outside the characterized range, so the study bounds that limit from below without locating it.

Step yield is unaffected by any parameter, with a predicted yield at the centre of the region of 96.1% and no factor moving it detectably, which is the expected result for a non-binding step and removes yield as a constraint on the operating region.

Two boundaries apply to all of this. The mechanistic account holds over the ranges given in Table 4.1 and at the qualified scale-down model (§3.1), and the models describe mean levels of the responses, with the variability around those means treated separately in §7 and §8.

6 Design space

The operating region for the anion exchange step is the intersection of two multivariate constraints: a viral clearance constraint set by load pH and load conductivity, and a pool purity constraint set by load pH and equilibration and wash-1 conductivity. Each holds over the ranges given in Table 4.1, with protein load and flow rate anywhere within their characterized ranges, and the narrower of the two constraints governs at any given operating point.

Consider first the viral clearance plane, whose worst case corner is low load pH with high load conductivity, since both effects act to reduce clearance and there is no interaction to soften the corner. At that corner the fitted model predicts an MVM log reduction factor of 3.53 $\log_{10}$ against the required step contribution of 3.28 $\log_{10}$, a margin of 0.25 $\log_{10}$. The same corner gives an XMuLV log reduction factor of 5.60 $\log_{10}$ against a required contribution of 3.78 $\log_{10}$. MVM is therefore the binding virus at every point in the region.

The pool purity plane has its worst case corner at low load pH with high equilibration and wash-1 conductivity, where the model predicts a pool HCP concentration of 31.9 ng/mg against an acceptance criterion of 100 ng/mg. Pool purity does not constrain the region anywhere inside the characterized ranges. Viral clearance is the constraint that defines the operating region for this step.

The normal operating ranges given in Table 4.1 sit inside the characterized region, which in turn sits inside the knowledge space bounded by the same table. At the worst case corner of the normal operating ranges the model predicts an MVM log reduction factor of 4.20 $\log_{10}$, a margin of 0.92 $\log_{10}$ over the requirement, while pool HCP at the corresponding purity corner is 22.0 ng/mg, and the commercial step will be operated well inside the region on both constraints.

An alternative approach was considered and rejected. The region could have been declared as a set of independent one-dimensional ranges, one per parameter, each proven with the others held at set-point. While that approach is simpler to state, it discards the information that load pH and load conductivity trade against each other. It unnecessarily constrains the operating region. The multivariate statement is used instead, and the univariate ranges are reported separately in §7.

Three bounds apply to the design space claim. The first is a range bound, since the region is defined only over the parameter ranges in Table 4.1 and no claim is made outside them. The second is a model bound, since the response-surface models predict mean levels of each response. At the edge of a region defined on mean levels the reliability of meeting the criterion approaches one half if the variability is symmetric about the mean. Assurance at commercial scale is established separately in §8 and is not read off this section. The third is a scale bound, since the region rests on the qualified scale-down model (§3.1), and design spaces are not confirmed at scale at the edges of the ranges.

Movement within the design space is not considered a change from a regulatory filing perspective, as stated in ICH Q8 (International Council for Harmonisation 2009), although any planned movement is still assessed prospectively under the pharmaceutical quality system in accordance with ICH Q10 (International Council for Harmonisation 2008). Operation outside the region requires change control and supporting data.

7 Proven acceptable ranges

Each characterized parameter is proven acceptable across its entire characterization range, for every quality attribute the step governs. The acceptance basis differs by attribute: for pool HCP the criterion is the drug substance specification, which is applied directly to the pool because no later step clears HCP, whereas for the two viral clearance responses the criterion is the step contribution back-calculated from the cumulative requirement, as described above (§2.2).

Two analyses were run for every combination of attribute and parameter. The first holds the other parameters at set-point and finds the range of the parameter over which the predicted response stays within acceptance. The second varies the other parameters within their normal operating ranges by Monte-Carlo simulation of the fitted response-surface model. The proven acceptable range is then the range over which the 95% predictive interval of the response stays within acceptance. Because the second analysis is a robustness criterion, it cannot be wider than the first. Both are given in Table 7.1.

Table 7.1: Proven acceptable ranges at set-point and with the remaining parameters varying within their normal operating ranges.

CQA	Parameter	Char. range	Unit	PAR (set-point)	PAR (NOR)
Pool HCP (ng/mg)	Load pH	7.2–7.8	pH	7.2–7.8	7.2–7.8
Pool HCP (ng/mg)	Equil/Wash-1 conductivity	1.6–3.6	mS/cm	1.6–3.6	1.6–3.6
Pool HCP (ng/mg)	Load conductivity	3–8	mS/cm	3–8	3–8
Pool HCP (ng/mg)	Protein load	50–300	g/L resin	50–300	50–300
XMuLV LRF ($\log_{10}$)	Load pH	7.2–7.8	pH	7.2–7.8	7.2–7.8
XMuLV LRF ($\log_{10}$)	Equil/Wash-1 conductivity	1.6–3.6	mS/cm	1.6–3.6	1.6–3.6
XMuLV LRF ($\log_{10}$)	Load conductivity	3–8	mS/cm	3–8	3–8
XMuLV LRF ($\log_{10}$)	Protein load	50–300	g/L resin	50–300	50–300
MVM LRF ($\log_{10}$)	Load pH	7.2–7.8	pH	7.2–7.8	7.2–7.8

CQA	Parameter	Char. range	Unit	PAR (set-point)	PAR (NOR)
MVM LRF ($\log_{10}$)	Equil/Wash-1 conductivity	1.6–3.6	mS/cm	1.6–3.6	1.6–3.6
MVM LRF ($\log_{10}$)	Load conductivity	3–8	mS/cm	3–8	3–8
MVM LRF ($\log_{10}$)	Protein load	50–300	g/L resin	50–300	50–300

The two analyses coincide in every row, and both equal the characterization range. No parameter has a narrower proven acceptable range under NOR co-variation than it has at set-point. The practical reading is that the step meets every acceptance criterion at any single parameter setting inside the ranges studied, provided the remaining parameters stay within their normal operating ranges.

That result is not a statement that any combination of extremes is acceptable, because both analyses vary one parameter to its edge while the others stay at set-point or within their normal operating ranges. The design space in §6 is the multivariate statement and it is the binding constraint on the operating region, and the worst case corner identified there lies outside the conditions covered by either PAR analysis, since it sets two parameters to their edges simultaneously.

The margins behind the table are not uniform, and the tightest case is the MVM log reduction factor against load pH, which is its governing parameter. Under NOR co-variation the lowest point of the 95% predictive interval anywhere in that range is 3.33 $\log_{10}$, which clears the required step contribution by 0.05 $\log_{10}$. The proportion of simulated results within acceptance never falls below 98.0% at any point in the range. Both analyses are shown in Figure 7.1.

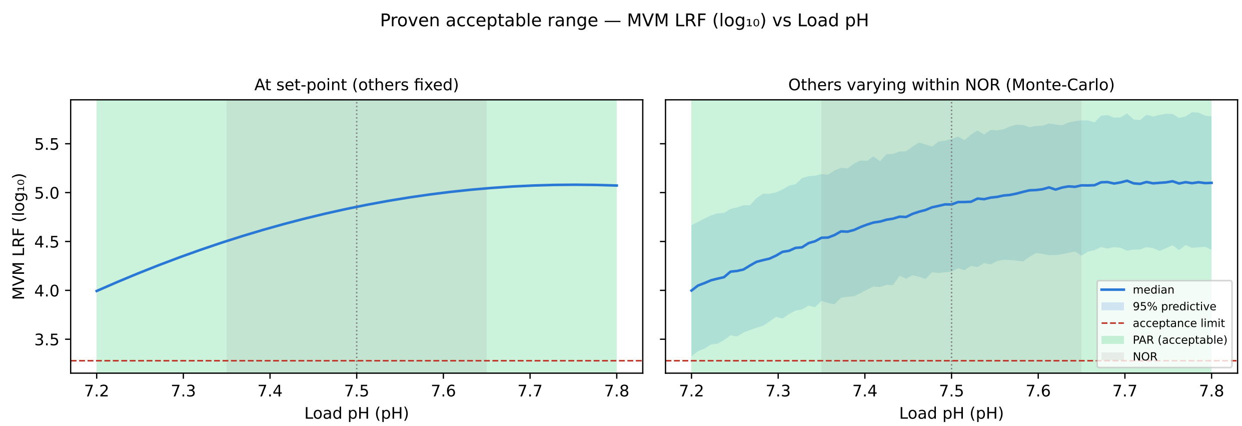

Figure 7.1: Proven acceptable range for the MVM log reduction factor against load pH. Left panel at set-point, right panel with the remaining parameters varying within their normal operating ranges.

The acceptable region is shaded in the figure and spans the whole characterized range in both panels. The predictive interval widens in the right panel, as expected once the other parameters are allowed to move. Its lower bound still stays above the acceptance line. The equivalent analysis for XMuLV clearance is shown in Figure 7.2.

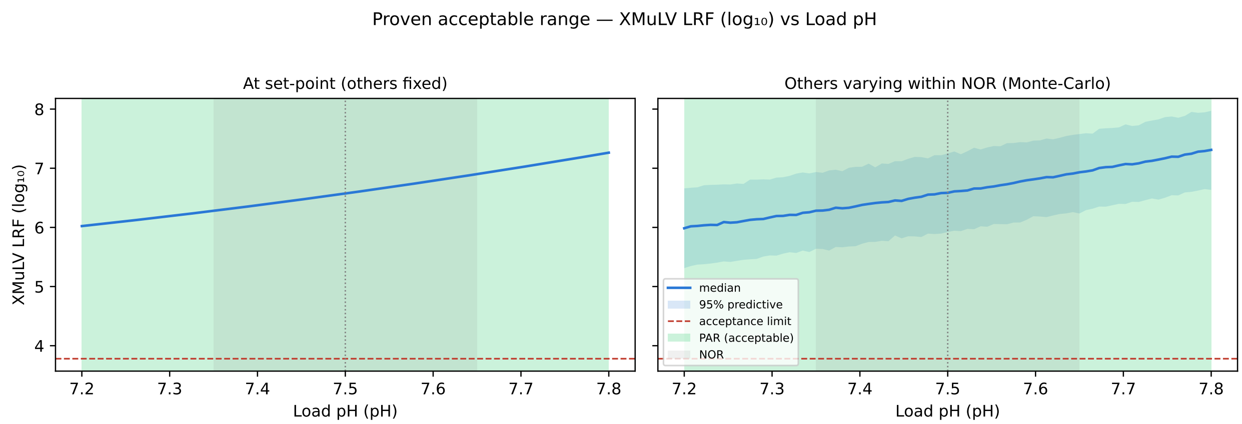

Figure 7.2: Proven acceptable range for the XMuLV log reduction factor against load pH. Left panel at set-point, right panel with the remaining parameters varying within their normal operating ranges.

XMuLV clearance clears its required step contribution by a wider margin at every point, which is why MVM governs the region. The analysis for pool HCP against equilibration and wash-1 conductivity, its governing parameter, is shown in Figure 7.3.

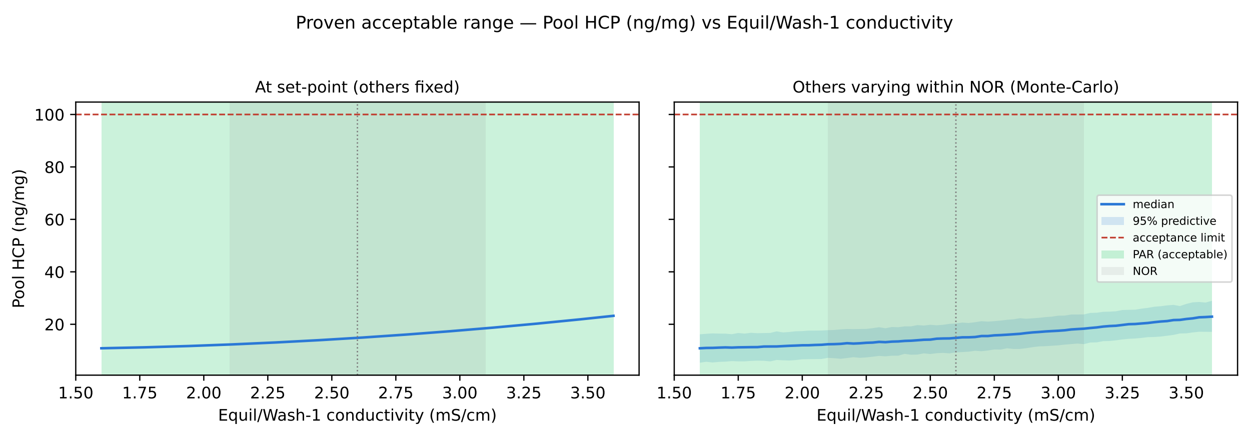

Figure 7.3: Proven acceptable range for pool host cell protein against equilibration/wash-1 conductivity. Left panel at set-point, right panel with the remaining parameters varying within their normal operating ranges.

The highest point reached by the upper predictive bound anywhere in that range is 29.0 ng/mg, which is well below the acceptance criterion of 100 ng/mg. Clearance of HCP at this step is supported by Protein A capture (Step 5) and by cation exchange (Step 7), which together remove the bulk of the load. The ranges given here apply to the feed that those two steps deliver, and the clearance achieved at each of them is reported in PCR-005 and PCR-007.

Two bounds apply to the proven acceptable ranges. They are computed from the fitted response-surface models and inherit the model bound stated in §6, and they are valid only over the characterization ranges, with no extrapolation beyond those ranges claimed.

8 Process capability and robustness

Commercial-scale capability was estimated by Monte-Carlo simulation of 2,000 batches with every parameter drawn within its normal operating range, and the result for each attribute the step governs is given in Table 8.1.

Table 8.1: Commercial-scale capability for the attributes governed by the step.

CQA	Crit.	Spec	Acceptance	Mean ± SD	Cpk
Host Cell Protein (HCP)	M-H	upper	0–100	15.2 ± 4.6	6.14
Residual DNA	VL	upper	0–0.001	0.0001 ± 0	10.8
Leached Protein A	L-M	upper	0–5	0.003 ± 0.0006	2.79e+03
Viral clearance — XMuLV (enveloped)	VH	lower	16.7–30	19.5 ± 0.61	1.53
Viral clearance — MVM (parvovirus)	VH	lower	8.6–20	10.4 ± 0.4	1.51

The MVM clearance claim carries the tightest capability in the table and in the whole drug substance process, with a capability index of 1.51 against a simulated mean of 10.38 $\log_{10}$ and a standard deviation of 0.40 $\log_{10}$. The margin from the simulated mean to the cumulative requirement is 1.78 $\log_{10}$. The XMuLV claim follows at 1.53 with a margin of 2.81 $\log_{10}$. Both exceed the value commonly applied as a minimum capability index for a critical characteristic. The minimum across all drug substance attributes is 1.51, which is the MVM figure.

The remaining attributes clear their limits by much wider margins, with pool HCP at 6.1 against a simulated mean of 15.2 ng/mg, and residual DNA and leached Protein A further still from their limits at 10.8 and 2785 respectively. Neither of those two constrains the process.

The MVM and XMuLV figures are cumulative process claims and not step claims, and the modular ledger behind them is given in Table 8.2.

Table 8.2: Modular viral clearance ledger for the drug substance process.

step	XMuLV	MVM
Low-pH Viral Inactivation	7.1	0
Anion Exchange (AEX)	5.95	4.71
Virus Filtration	5.82	5.32

step	XMuLV	MVM
Cumulative	18.9	10

Anion exchange contributes 4.71 $\log_{10}$ of the cumulative MVM figure of 10.03 $\log_{10}$ (47% of the total), and small-virus retentive filtration contributes the remaining 5.32 $\log_{10}$ as reported in PCR-009. The low-pH hold is not credited with any MVM clearance, because inactivation of a non-enveloped parvovirus at low pH is not reliable. For XMuLV the step contributes 5.95 $\log_{10}$ of a cumulative 18.87 $\log_{10}$, alongside 7.10 $\log_{10}$ from the low-pH hold reported in PCR-006 and 5.82 $\log_{10}$ from filtration. The three mechanisms are independent, which is what allows the contributions to be added, and the consolidated claim is made in PCMR-001.

Two bounds apply to the capability figures. They describe operation within the normal operating ranges and not within the wider design space, and they rest on the scale-down model, on the small-scale spiking model, and on the assumption that commercial-scale variability is captured by the simulated variance components.

9 Parameter classification

All 5 characterized parameters are classified as well-controlled critical process parameters under SOP-4001, because each of them is either linked to a critical quality attribute or bounds the conditions under which a clearance claim is made. Each also has a low risk of leaving the design space, since the region is wide relative to the control capability of the equipment. No parameter at this step is classified as a critical process parameter, and none is classified as a key process parameter or a general process parameter. The reasoning for each is given below.

- Load pH is a WC-CPP because it has the largest effect on both viral clearance responses (Table 5.4, Table 5.5) and the second largest on pool HCP. It is controlled by in-line titration with a verified probe, and its normal operating range is narrow relative to the characterized range, so the risk of leaving the region is low.
- Equilibration and wash-1 conductivity is a WC-CPP because it is the governing parameter for pool HCP and interacts with load pH, although it has no significant effect on either clearance response. Buffer conductivity is set at preparation and verified at release under SOP-2103, so it is readily controlled.
- Load conductivity is a WC-CPP because it is the second largest effect on both clearance responses and defines one axis of the viral clearance constraint described in §6. It is set by the adjustment of the cation exchange pool before loading, and it is verified before the load is applied.
- Protein load is a WC-CPP even though it had no significant effect on any response over a range extending to 300 g/L resin (§5.2), because the viral clearance claim is supported only up to the maximum load at which clearance was demonstrated. The class therefore reflects the bound on the claim and not a demonstrated effect, and load is measured directly from the load mass and the packed bed volume.
- Operating flow rate is a WC-CPP on the same bounding logic, since it showed no effect in the univariate assessment across its characterized range and the spiking runs that support the clearance claim were executed at the maximum flow rate. Flow rate is controlled by the chromatography skid (SOP-2011) and is recorded continuously.

The two null results are retained in the knowledge space and are not discarded, because protein load and flow rate are evidence that the step is robust to load and to residence time over the ranges studied. That robustness simplifies the control strategy, since only the three chemistry parameters (load pH, load conductivity, and equilibration and wash-1 conductivity) have to be held closely in order to govern the attributes this step controls.

For context across the whole train, the parameter counts by class are 20 WC-CPP, 1 CPP, 10 KPP and 6 GPP, and the consolidated classification is presented in PCMR-001.

10 Contribution to the control strategy

The anion exchange step contributes four elements to the drug substance control strategy.

The first is parameter control. Load pH, load conductivity and equilibration and wash-1 conductivity are held within the normal operating ranges given in Table 4.1, which lie inside the design space defined in §6 with the margins stated there. Protein load and flow rate are held within their normal operating ranges, which bound the conditions under which the viral clearance claim was demonstrated.

The second is in-process monitoring. Load pH and load conductivity are verified on the adjusted load before the column is loaded, equilibration and wash-1 buffer conductivity and pH are verified at buffer release under SOP-2103, and pool HCP is measured on every batch and trended against the capability reported below (§8). Step yield is monitored as a process performance indicator, and §5.2 shows that it is not quality-linked at this step.

The third is life-cycle control. Column packing quality, sanitization and resin lifetime are controlled under SOP-2008, and column efficiency is monitored between runs by transitional analysis, with repacking triggered on the criteria in that procedure. Resin life-cycle behaviour is not characterized in this study, which was executed on a single resin lot within each design.

The fourth is a control arising from a deviation. The load material presented to this step must be representative of the commercial cation exchange pool, and an extended neutralized hold of that pool is the condition that made it unrepresentative in the first execution (§13.1). A hold time limit on the neutralized cation exchange pool is therefore carried into the Stage 2 control strategy, and the acidic charge variant content of the load is verified by imaged capillary isoelectric focusing (AMV-3013) before the step is run.

Two limits on this contribution are stated explicitly. The step does not assure viral safety on its own, since the cumulative MVM claim requires small-virus retentive filtration and the cumulative XMuLV claim requires the low-pH hold as well (§8, Table 8.2). The step also makes no contribution to the control of glycan variants, charge variants or aggregate, which are governed at Step 3 and Step 7. All contributions are consolidated in PCMR-001.

11 Discussion

The step is well understood over the ranges studied, in that three parameters govern the attributes it controls and the mechanism behind each of them is established rather than inferred. Viral clearance is a charge partitioning process governed by load pH and load conductivity, while pool purity is governed by the ionic environment of the column during flow-through and therefore by the equilibration and wash-1 conductivity. Two parameters are inactive over wide ranges. The resulting operating region is bounded by viral clearance and not by pool purity. The binding attribute inside that region is MVM clearance.

The scale-down model is the foundation of every claim in this report, so its representativeness deserves a direct statement. The model reproduces the commercial step at target conditions on the input and output attributes compared during qualification, but it is not independently qualified at the edges of the characterized ranges, and design spaces are not confirmed at scale at the edges of the ranges. The verification runs (§13.2) provide the only direct evidence at corner conditions, and they too were run at small scale.

Three deviations affected the campaign, and the most serious of them invalidated the first execution of both designs. The load used in that execution carried an acidic charge variant content of 33.5% against 21.0% in the requalified material, and pool HCP reached 150 ng/mg at one corner of that design, so both designs were re-executed in full on requalified load. The requalified data show no trace of the anomalous interaction seen in the first execution, which confirms the root cause. The other two deviations were a collection criterion correction and a buffer pH excursion on a single screening run, and none of the three changed a classification or a region boundary (§13).

Four limitations are recorded. The first is the mean-level nature of the response-surface models, which is why assurance is established by simulation (§8) and is not read off the contours. The second is the precision of the HCP assay, at 9.5% relative standard deviation, which sets a floor on the size of an effect that this study could have resolved for that response. The third is resin life-cycle, which is controlled procedurally and is not characterized here. The fourth is the small-scale spiking model, which is qualified as representative but is not a commercial-scale measurement, in common with all modular viral clearance claims made under ICH Q5A (International Council for Harmonisation 2023a).

No parameter at this step required the most stringent control class, because the design space is wide relative to the control capability of the equipment for all five parameters, and each of them is either measured directly and continuously (pH, conductivity, flow rate) or verified before use. That combination is what distinguishes a well-controlled critical process parameter from a critical process parameter under SOP-4001.

12 Conclusions

The anion exchange step is well characterized and robust over the ranges studied.

A multivariate operating region is defined over load pH, load conductivity and equilibration and wash-1 conductivity, with protein load and flow rate free within their characterized ranges, and at the worst corner of that region the predicted MVM log reduction factor is 3.53 $\log_{10}$ against a required step contribution of 3.28 $\log_{10}$. Every characterized parameter is proven acceptable across its full characterization range, both at set-point and with the other parameters varying within their normal operating ranges.

At commercial scale the governed attributes meet their acceptance criteria, and the tightest capability is the cumulative MVM claim at 1.51, which is also the minimum across the drug substance process. All 5 characterized parameters are classified as well-controlled critical process parameters, and no critical process parameter and no key process parameter is assigned at this step.

Three deviations were recorded, investigated and resolved, one of which invalidated the first execution of both designs, and those designs were re-executed in full on requalified load material. No deviation altered a parameter classification or a boundary of the operating region.

The step does not assure viral safety on its own, and it makes no claim for glycan variants, charge variants or aggregate. Its contribution is consolidated with the other steps in PCMR-001.

13 Deviations from the plan

Three deviations from PCP-008 were recorded during execution. Each was investigated, assigned a root cause, impact assessed and dispositioned before the data were used. One of them invalidated the first execution of both designs and forced a full re-execution. The register is given in Table 13.1.

Table 13.1: Register of deviations recorded during execution of PCP-008.

Deviation	Summary	Detected during	Disposition
DEV-008-01	Non-representative (deamidated) load in the first DoE execution; designs invalidated and re-executed	load-material charge-variant verification by icIEF (AMV-3013)	invalidated_and_re_executed
DEV-008-02	Trailing-edge UV pool-collection set-point slightly permissive in both executions	post-execution pool-fraction reconciliation (AMV-3012)	corrected_by_modelling_and_verification_runs
DEV-008-03	Equilibration/Wash-1 buffer released below its pH acceptance range on one screening run	buffer release testing per SOP-2103	retained

The three differ considerably in severity: the first invalidated a complete execution of the study, the second applied to both executions and was corrected by modelling and by verification runs, and the third affected one screening run and was retained. Each is given in turn below.

13.1 Non-representative load material in the first execution

Pool HCP in the first execution of the screening design reached 150 ng/mg at the corner combining high equilibration and wash-1 conductivity with high protein load, which is above the acceptance criterion of 100 ng/mg applied to the pool. The corresponding maximum in the re-executed design is 36 ng/mg.

The investigation began with the load material, because the excursion appeared at a load corner and not at a chemistry corner. Charge variant verification of the load by imaged capillary isoelectric focusing (AMV-3013) found an acidic variant content of 33.5% in lot LOT-CEX-8842, against 21.0% in the requalified load used for the reported execution. The root cause was an extended neutralized hold of the cation exchange pool before adjustment, lasting 36 hours, which promoted asparagine deamidation. Because deamidation lowers the isoelectric point of the affected species, the additional acidic product species compete with HCP for the available binding sites on a strong anion exchanger operated in flow-through mode. That competition scales with the mass of product applied, which is why the anomaly presented as an interaction between protein load and wash conductivity.

The affected screening and response-surface designs were invalidated for the purpose of defining the operating region, and both were re-executed in full on a requalified, representative load (LOT-CEX-8851). The first execution is retained as a controlled record and is referenced here only to confirm the root cause. It is not used in any analysis in this report. The three terms that carry the anomaly are compared in Table 13.2.

Table 13.2: Screening effects on pool host cell protein in the invalidated first execution and in the re-executed study.

Term	Effect (superseded)	p (superseded)	Effect (requalified)	p (requalified)
B	50.4	0.000195	12.1	9.48e-06
D	35.6	0.00181	0.885	0.491
BD	34.8	0.00211	0.224	0.86

The interaction between protein load and equilibration and wash-1 conductivity is 34.8 ng/mg in the first execution ($p = 0.002$), whereas in the re-executed study the same term is 0.22 ng/mg ($p = 0.860$). The protein load main effect falls from 35.6 ng/mg to 0.89 ng/mg. Both terms are absent from the requalified data, which is the result the deamidation mechanism predicts and which confirms the root cause: the load material caused the excursion and the operating parameters did not.

The corrective action is a hold time limit on the neutralized cation exchange pool, together with verification of the acidic charge variant content of the load before the anion exchange step is run, and both are carried into the Stage 2 control strategy (§10).

13.2 Collection criterion on the descending edge

The product pool was collected to a descending UV absorbance threshold of 180 mAU in both executions, where SOP-2011 specifies 120 mAU. Since collecting to the lower threshold extends the pool further down the descending edge, the executed runs collected a narrower pool than the commercial step will collect, so the pool HCP concentrations reported in §5 apply to a pool boundary that is not the commercial one and may understate the commercial value.

The discrepancy was found during post-execution reconciliation of the pool fractions against the chromatograms and against the HCP results measured under AMV-3012, and the root cause was an incorrect collection set-point carried in the laboratory method file. The set-point was corrected to 120 mAU.

The impact was assessed in two ways. The additional trailing-edge material was quantified from the recorded chromatograms and added to the measured pool composition. Verification runs at the corrected set-point followed, 3 in total, of which 2 were placed at corners of the operating region and not at the centre. All verification results fell inside the region defined in §6 and within the acceptance criterion for pool HCP. The relative 95% confidence half-width of the verification set is 23.6%, which follows from the intermediate precision of the HCP assay at that number of runs. The verification demonstrates that the region holds at the corrected collection criterion. It does not resolve differences in pool HCP smaller than that half-width.

The collection criterion is fixed at 120 mAU in SOP-2011 for Stage 2. No parameter classification and no boundary of the operating region was changed.

13.3 Equilibration and wash-1 buffer pH excursion

One lot of equilibration and wash-1 buffer (LOT-BUF-8815) was released at pH 7.32, against a lower acceptance limit of 7.40 and a target of 7.50, and that lot was used on one screening run. The root cause was a titration endpoint error at buffer preparation (SOP-2103), identified during release testing under SOP-2103.

Three arguments bound the impact. Buffer pH is not a factor in either design, and the load pH factor was set and verified independently of the buffer. The measured value lies inside the characterized range for load pH, which spans 7.2 to 7.8, so the affected run remains inside the knowledge space. The affected run belongs to the screening design, which is not the predictive model, whereas the operating region in §6 and the proven acceptable ranges in §7 rest on the response-surface design, which was executed with released buffer throughout.

The deviation was retained with no re-execution, and buffer preparation under SOP-2103 now requires a second verification of the titration endpoint before release.

Appendices A–D

Appendix A — Screening design matrix and responses

The coded screening design matrix with its measured responses is given below, with factor letters following Table 4.2 and coded settings taking the edges and the centre of each characterized range.

Table 13.3: Coded screening design matrix with measured responses.

Run	Type	A	B	C	D	Pool HCP (ng/mg)	XMuLV LRF (log_{10})	MVM LRF (log_{10})	Step yield
1	facto-rial	-1	-1	-1	-1	12.5	6.16	4.59	0.99
2	facto-rial	-1	-1	-1	1	15.6	6.15	4.43	0.975
3	facto-rial	-1	-1	1	-1	12.1	5.47	3.76	0.951
4	facto-rial	-1	-1	1	1	14.6	5.56	3.96	0.964
5	facto-rial	-1	1	-1	-1	35.7	6.34	4.74	0.965
6	facto-rial	-1	1	-1	1	35	6.3	4.83	0.995
7	facto-rial	-1	1	1	-1	29.2	5.92	4.01	0.974
8	facto-rial	-1	1	1	1	29.8	5.95	3.94	0.969
9	facto-rial	1	-1	-1	-1	9.62	7	5.91	0.973
10	facto-rial	1	-1	-1	1	7.37	7.4	5.61	0.953
11	facto-rial	1	-1	1	-1	8.69	6.61	5.25	0.972
12	facto-rial	1	-1	1	1	8.04	6.71	4.89	0.973
13	facto-rial	1	1	-1	-1	12.7	6.59	5.91	0.995
14	facto-rial	1	1	-1	1	15	8.34	5.97	0.981

Run	Type	A	B	C	D	Pool HCP (ng/mg)	XMuLV LRF (log_{10})	MVM LRF (log_{10})	Step yield
15	facto-rial	1	1	1	-1	12.9	6.69	5.15	0.966
16	facto-rial	1	1	1	1	15	6.52	5.31	0.965
17	center	0	0	0	0	16.2	6.42	4.54	0.966
18	center	0	0	0	0	17.5	6.8	5.08	0.975
19	center	0	0	0	0	11.6	5.95	4.83	0.974

Appendix B — Response-surface design matrix and responses

The coded response-surface design matrix with its measured responses is given below, in which the run types are factorial, axial and centre.

Table 13.4: Coded response-surface design matrix with measured responses.

Run	Type	A	B	C	D	Pool HCP (ng/mg)	XMuLV LRF (log_{10})	MVM LRF (log_{10})	Step yield
1	facto-rial	-1	-1	-1	-1	13.5	6.72	4.55	0.951
2	facto-rial	-1	-1	-1	1	17.3	6.03	5.21	0.989
3	facto-rial	-1	-1	1	-1	14.1	5.87	4.02	0.995
4	facto-rial	-1	-1	1	1	15	5.3	3.76	0.951
5	facto-rial	-1	1	-1	-1	28.5	6.53	4.77	0.988
6	facto-rial	-1	1	-1	1	25.5	6.71	4.71	0.963
7	facto-rial	-1	1	1	-1	31.3	5.36	3.91	0.995
8	facto-rial	-1	1	1	1	32.3	5.06	3.36	0.966
9	facto-rial	1	-1	-1	-1	5.66	7.5	6.14	0.974
10	facto-rial	1	-1	-1	1	8.37	7.83	5.61	0.989
11	facto-rial	1	-1	1	-1	7.56	6.49	4.54	0.972
12	facto-rial	1	-1	1	1	7.43	7.13	4.68	0.988
13	facto-rial	1	1	-1	-1	17.6	7.18	6.2	0.974

Run	Type	A	B	C	D	Pool HCP (ng/mg)	XMuLV LRF ($\log_{10}$)	MVM LRF ($\log_{10}$)	Step yield
14	facto-rial	1	1	-1	1	18	8.12	6.33	0.968
15	facto-rial	1	1	1	-1	13.9	6.58	4.53	0.96
16	facto-rial	1	1	1	1	13.9	6.37	5.16	0.973
17	axial	-1	0	0	0	23	5.83	4.1	0.971
18	axial	1	0	0	0	12	7.37	4.91	0.965
19	axial	0	-1	0	0	8.1	6.42	4.93	0.946
20	axial	0	1	0	0	27.5	6	5.02	0.972
21	axial	0	0	-1	0	15.6	7.44	5.14	0.952
22	axial	0	0	1	0	15.5	5.93	4.56	0.967
23	axial	0	0	0	-1	13	6.52	4.81	0.968
24	axial	0	0	0	1	14.3	6.7	5.11	0.952
25	center	0	0	0	0	15	6.59	4.42	0.958
26	center	0	0	0	0	10.7	6.03	5.2	0.975
27	center	0	0	0	0	16.3	7.05	5.04	0.977
28	center	0	0	0	0	12.6	6.86	4.92	0.953

Appendix C — Full effect and coefficient tables

Complete screening effect tables for all responses follow, and then the response-surface coefficient tables for the responses not tabulated in §5.

Pool HCP (ng/mg)

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
B	12.1	6.04	0.613	9.85	9.48e-06	***
A	-11.9	-5.96	0.613	-9.72	1.05e-05	***
AB	-6.63	-3.32	0.613	-5.41	0.000642	***
C	-1.66	-0.829	0.613	-1.35	0.213	
AC	1.65	0.827	0.613	1.35	0.214	
BC	-1.21	-0.605	0.613	-0.986	0.353	
D	0.885	0.443	0.613	0.722	0.491	
AD	-0.498	-0.249	0.613	-0.406	0.696	
CD	0.261	0.13	0.613	0.212	0.837	
BD	0.224	0.112	0.613	0.182	0.86	

XMuLV LRF ($\log_{10}$)

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
A	1	0.501	0.0982	5.1	0.000934	***
C	-0.605	-0.302	0.0982	-3.08	0.0151	*
D	0.268	0.134	0.0982	1.37	0.209	
CD	-0.255	-0.128	0.0982	-1.3	0.23	
AD	0.25	0.125	0.0982	1.27	0.24	
B	0.198	0.0991	0.0982	1.01	0.342	
BD	0.126	0.0632	0.0982	0.643	0.538	
AC	-0.0968	-0.0484	0.0982	-0.493	0.635	
AB	-0.094	-0.047	0.0982	-0.478	0.645	
BC	-0.0169	-0.00844	0.0982	-0.0859	0.934	

MVM LRF ($\log_{10}$)

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
A	1.22	0.61	0.0446	13.7	7.83e-07	***
C	-0.715	-0.357	0.0446	-8.01	4.32e-05	***
B	0.182	0.091	0.0446	2.04	0.0757	
BD	0.107	0.0536	0.0446	1.2	0.263	
AD	-0.0627	-0.0314	0.0446	-0.704	0.502	
D	-0.0451	-0.0226	0.0446	-0.506	0.626	
BC	-0.0442	-0.0221	0.0446	-0.496	0.634	
CD	0.0334	0.0167	0.0446	0.375	0.718	
AC	0.0135	0.00677	0.0446	0.152	0.883	
AB	-0.0132	-0.00659	0.0446	-0.148	0.886	

Step yield

Table 13.5: Complete screening effect estimates for every measured response.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
C	-0.0116	-0.00582	0.00315	-1.85	0.102	
B	0.00743	0.00371	0.00315	1.18	0.272	
AD	-0.00715	-0.00357	0.00315	-1.13	0.29	
AC	0.00513	0.00257	0.00315	0.815	0.439	
BC	-0.00392	-0.00196	0.00315	-0.622	0.551	
BD	0.00383	0.00192	0.00315	0.608	0.56	
CD	0.00346	0.00173	0.00315	0.549	0.598	
AB	0.00141	0.000707	0.00315	0.224	0.828	
D	-0.00136	-0.000679	0.00315	-0.215	0.835	
A	-0.000663	-0.000332	0.00315	-0.105	0.919	

XMuLV LRF ($\log_{10}$)

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	6.57	0.109	60.6	2.51e-17	***
A	0.621	0.074	8.39	1.33e-06	***
B	-0.076	0.074	-1.03	0.323	
C	-0.554	0.074	-7.48	4.6e-06	***
D	0.0287	0.074	0.388	0.704	
AB	-0.0278	0.0785	-0.355	0.729	
AC	0.0223	0.0785	0.285	0.78	
AD	0.192	0.0785	2.44	0.0295	*
BC	-0.118	0.0785	-1.5	0.158	
BD	0.0577	0.0785	0.735	0.475	
CD	-0.0751	0.0785	-0.957	0.356	
A^2	0.0696	0.195	0.356	0.727	
B^2	-0.328	0.195	-1.68	0.118	
C^2	0.152	0.195	0.779	0.45	
D^2	0.0761	0.195	0.39	0.703	

Step yield

Table 13.6: Response-surface coefficients for the responses not tabulated in §5.

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	0.961	0.00501	192	7.92e-24	***
A	-0.000294	0.00342	-0.086	0.933	
B	0.00022	0.00342	0.0643	0.95	
C	0.00109	0.00342	0.318	0.756	
D	-0.00203	0.00342	-0.594	0.563	
AB	-0.00464	0.00362	-1.28	0.223	
AC	-0.00175	0.00362	-0.482	0.637	
AD	0.00608	0.00362	1.68	0.117	
BC	-0.000152	0.00362	-0.042	0.967	
BD	-0.00433	0.00362	-1.2	0.253	
CD	-0.00413	0.00362	-1.14	0.275	
A^2	0.00932	0.00902	1.03	0.32	
B^2	0.000524	0.00902	0.0581	0.955	
C^2	0.00152	0.00902	0.168	0.869	
D^2	0.00177	0.00902	0.196	0.847	

Appendix D — Analytical methods summary

Validated method performance for the methods applied at this step is given below, and the remaining methods listed in Table 3.1 are covered by their own validation reports.

Table 13.7: Performance of the validated analytical methods applied at this step.

Method	Analytical method	Intermediate precision	LOQ	LOQ unit	Accuracy (%)	Assay share of variance
AMV-3016	Leached Protein A by ELISA	6.5	0.25	ppm	97.4	0.41
AMV-3017	XMuLV Infectivity Titre (TCID50)	0.3	1.5	log10 TCID50/mL	96	0.28
AMV-3018	MVM Infectivity Titre (TCID50/qPCR)	0.32	1.4	log10 TCID50/mL	96	0.3
AMV-3012	Host-Cell Protein by ELISA	9.5	1	ng/mg	95	0.4

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